$$
C_{uu}^{i1_i2_i3_i4_} \to
$$

$$
\frac{1}{16\,\pi^2}\Big(\frac{1}{18}\,g_3^4\,(3\,\delta_{i1i4}\,\delta_{i2i3}-\delta_{i1i2}\,\delta_{i3i4})\,LF_{3,0}[m_3]+\frac{1}{12}\,g_3^4\,(3\,\delta_{i1i4}\,\delta_{i2i3}-\delta_{i1i2}\,\delta_{i3i4})\,LF_{4,-1}[m_3]+
$$

$$
\frac{4}{45}\,g_3^4\,(-3\,\delta_{i1i4}\,\delta_{i2i3}+\delta_{i1i2}\,\delta_{i3i4})\,LF_{5,-2}[m_3]+
$$

$$
\frac{1}{486}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(16\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,0}\left[m_{\tilde{d}}^{p}\right]-
$$

$$
\frac{5}{1296}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(16\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,-1}\left[m_{\tilde{d}}^{p}\right]+
$$

$$
\frac{2}{1215}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(16\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{5,-2}\left[m_{\tilde{d}}^{p}\right]+
$$

$$
\frac{8}{81}\sum_p\,g_1^4\,LF_{3,0}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{5}{27}\sum_p\,g_1^4\,LF_{4,-1}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{32}{405}\sum_p\,g_1^4\,LF_{5,-2}\left[m_{\tilde{e}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{4}{81}\sum_p\,g_1^4\,LF_{3,0}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}-
$$

$$
\frac{5}{54}\sum_p\,g_1^4\,LF_{4,-1}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{16}{405}\sum_p\,g_1^4\,LF_{5,-2}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{1}{243}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(4\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,0}\left[m_{\tilde{q}}^{p}\right]-
$$

$$
\frac{5}{648}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(4\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,-1}\left[m_{\tilde{q}}^{p}\right]+
$$

$$
\frac{4}{1215}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(4\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{5,-2}\left[m_{\tilde{q}}^{p}\right]+
$$

$$
\frac{1}{486}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(64\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,0}\left[m_{\tilde{u}}^{p}\right]-
$$

$$
\frac{5}{1296}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(64\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,-1}\left[m_{\tilde{u}}^{p}\right]+
$$

$$
\frac{2}{1215}\sum_p\,\left(27\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+\left(64\,g_1^4-9\,g_3^4\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{5,-2}\left[m_{\tilde{u}}^{p}\right]+
$$

$$
\frac{4}{81}\,g_1^4\,LF_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{27}\,g_1^4\,LF_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{405}\,g_1^4\,LF_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_1,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_1,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_1,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_1,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_1,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_1,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_1,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_1,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_1,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_1,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_1,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_1,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_1,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_1,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_1,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_1,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_3,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_3,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{1296}\,\left(-225\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(-128\,g_1^2+75\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_3,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{162}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_3,\,m_{\tilde{u}}^{i1}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_3,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_3,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{1296}\,\left(-225\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(-128\,g_1^2+75\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_3,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{162}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_3,\,m_{\tilde{u}}^{i2}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_3,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_3,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{1296}\,\left(-225\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(-128\,g_1^2+75\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_3,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{162}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_3,\,m_{\tilde{u}}^{i3}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_3,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,2,-1}\left[m_3,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{1296}\,\left(-225\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(-128\,g_1^2+75\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_3,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{162}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{4,1,-2}\left[m_3,\,m_{\tilde{u}}^{i4}\right]+
$$

$$
\frac{1}{54}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(-2\,g_1^2+3\,g_3^2\right)\delta_{i1i2}-9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(-9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(-2\,g_1^2+3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{2,1,0}\left[m_q^{p},\,\tilde{\mu}\right]+
$$

$$
\frac{1}{108}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i1i2}+9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{2,2,-1}\left[m_{\tilde{q}}^{p},\,\tilde{\mu}\right]+
$$

$$
\frac{1}{108}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i1i2}+9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{3,1,-1}\left[m_{\tilde{q}}^{p},\,\tilde{\mu}\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i1},\,m_1\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i1},\,m_1\right]+
$$

$$
\frac{1}{648}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}-g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i1},\,m_3\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i1},\,m_3\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i2},\,m_1\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i2},\,m_1\right]+
$$

$$
\frac{1}{648}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}-g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i2},\,m_3\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i2},\,m_3\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i3},\,m_1\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i3},\,m_1\right]+
$$

$$
\frac{1}{648}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}-g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i3},\,m_3\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i3},\,m_3\right]+
$$

$$
\frac{1}{243}\,g_1^2\,\left(-9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(-8\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i4},\,m_1\right]+
$$

$$
\frac{1}{486}\,g_1^2\,\left(9\,g_3^2\,\delta_{i1i4}\,\delta_{i2i3}+\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i4},\,m_1\right]+
$$

$$
\frac{1}{648}\,\left(9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}-g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{2,1,0}\left[m_{\tilde{u}}^{i4},\,m_3\right]+
$$

$$
\frac{1}{1296}\,\left(-9\,g_3^4\,\delta_{i1i4}\,\delta_{i2i3}+g_3^2\,\left(64\,g_1^2+3\,g_3^2\right)\delta_{i1i2}\,\delta_{i3i4}\right)LF_{3,1,-1}\left[m_{\tilde{u}}^{i4},\,m_3\right]+
$$

$$
\frac{1}{108}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i1i2}+9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(2\,g_1^2-3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{2,1,0}\left[\tilde{\mu},\,m_q^{p}\right]+
$$

$$
\frac{1}{54}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(-11\,g_1^2+3\,g_3^2\right)\delta_{i1i2}-9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(-9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(-11\,g_1^2+3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{3,1,-1}\left[\tilde{\mu},\,m_q^{p}\right]+
$$

$$
\frac{1}{108}\,\Big(\overline{y_u}^{pi3}\,\left(y_u^{pi4}\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i1i2}+9\,g_3^2\,y_u^{pi2}\,\delta_{i1i4}\right)+
$$

$$
\overline{y_u}^{pi1}\,\left(9\,g_3^2\,y_u^{pi4}\,\delta_{i2i3}+y_u^{pi2}\,\left(8\,g_1^2-3\,g_3^2\right)\delta_{i3i4}\right)\Big)\,LF_{4,1,-2}\left[\tilde{\mu},\,m_q^{p}\right]+
$$

$$
\frac{4}{81}\,g_1^4\,LF_{2,1,1,-1}\left[m_1,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{81}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\left[m_1,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{4}{81}\,g_1^4\,LF_{2,1,1,-1}\left[m_1,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{81}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\left[m_1,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{7}{48}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i2}\right]\delta_{i1i4}\,\delta_{i2i3}-\frac{1}{12}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i2}\right]\delta_{i1i4}\,\delta_{i2i3}+
$$

$$
\frac{1}{144}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{36}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{1}{144}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{36}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{7}{48}\,g_3^4\,LF_{2,1,1,-1}\left[m_3,\,m_{\tilde{u}}^{i3},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i4}\,\delta_{i2i3}-\frac{1}{12}\,g_3^4\,m_3^2\,LF_{2,1,1,0}\left[m_3,\,m_{\tilde{u}}^{i3},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i4}\,\delta_{i2i3}+
$$

$$
\frac{1}{4}\,\overline{y_u}^{pi1}\,\overline{y_u}^{ri3}\,y_u^{pi2}\,y_u^{ri4}\,LF_{2,1,1,-1}\left[\tilde{\mu},\,m_q^{p},\,m_q^{r}\right]+
$$

$$
\frac{1}{9}\,g_1^2\,g_3^2\,LF_{1,1,1,1,-1}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i2}\right]\delta_{i1i4}\,\delta_{i2i3}+
$$

$$
\frac{2}{9}\,m_1\,m_3\,g_1^2\,g_3^2\,LF_{1,1,1,1,0}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i2}\right]\delta_{i1i4}\,\delta_{i2i3}-
$$

$$
\frac{1}{27}\,g_1^2\,g_3^2\,LF_{1,1,1,1,-1}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-
$$

$$
\frac{2}{27}\,m_1\,m_3\,g_1^2\,g_3^2\,LF_{1,1,1,1,0}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i1},\,m_{\tilde{u}}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-
$$

$$
\frac{1}{27}\,g_1^2\,g_3^2\,LF_{1,1,1,1,-1}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-
$$

$$
\frac{2}{27}\,m_1\,m_3\,g_1^2\,g_3^2\,LF_{1,1,1,1,0}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i2},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+
$$

$$
\frac{1}{9}\,g_1^2\,g_3^2\,LF_{1,1,1,1,-1}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i3},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i4}\,\delta_{i2i3}+
$$

$$
\frac{2}{9}\,m_1\,m_3\,g_1^2\,g_3^2\,LF_{1,1,1,1,0}\left[m_1,\,m_3,\,m_{\tilde{u}}^{i3},\,m_{\tilde{u}}^{i4}\right]\delta_{i1i4}\,\delta_{i2i3}\Big)
$$